AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TT284
Web Technologies

Midterm Examination (MTA) – Makeup
Spring Semester 2024/2025

Date: 22/April/2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1

MCQ questions (10 Marks) (2 marks each)

1) The Advanced Research Projects Agency Network project of 1969 was aimed at creating _____.
   A. computer networks that would be resistant to intruders. It led to the creation of the Transfer Control Protocol and the Internet Protocol.
   B. computer networks that would be resistant to failure. It led to the creation of the Transmission Control Protocol and the Internet Protocol.
   C. protection protocols that would be resistant to intruders. It led to the creation of the HTTP Secure protocol.
   D. computer transactions that would be resistant to failure. It led to the creation of Web Security Protocols.

2) Hypertext Transfer Protocol Secure is _____.
   A. a version of the Hypertext Transfer Protocol combined with SSL/TLS protocol that provides encrypted communication and secure identification of a network web server.
   B. a version of the Hypertext Transfer Protocol combined with TCP/IP protocol that provides encrypted communication and secure identification of a network web server.
   C. a version of the Hypertext Transfer Protocol combined with FTP protocol that provides encrypted communication and secure identification of a network web server.
   D. a version of the Hypertext Transfer Protocol combined with SSL/TLS protocol that provides digital certificates to secure the communication between the client and the servers.

3) A regular expression is _____.
   A. a text string that represents a pattern for other objects and data types that can be validated to check that the output value is meeting the criteria that set by analyst of the application. The pattern could be characters, numbers, date, time, image, sound, and video.
   B. a regular instruction in the computer language to validate all inputs and outputs from the user based on predefined criteria such as the data type, size, identifier, and length of the input.
   C. a text string that represents a pattern for other strings that either match or do not match the pattern. The pattern is written using characters (i.e. literals), meta-characters and wildcards.
   D. a regular instruction in the computer language that represents a pattern for other instructions, the pattern is written using characters, meta-characters, mathematical operators, relational operators, and wildcards.

4) We might define 'surname' to be a key, in databases, this might not be a good option as there may be many individuals with the same surname. If you were to search a telephone book, for example, you would use the person's surname and then their first initial, which could be a _____ key.
   A. Double.
   B. primary.
   C. secondary.
   D. search.

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5) In the following code:
if (!$connection) {
Print ("Internal error". mysql_errno()); }

A. The code has a conditional expression '!' (i.e. not) connection that will be true if 'connection' evaluates to false. In this case an error message is printed and the 'mysql_error()' will be called.
B. The code has a conditional expression '!' (i.e. not) connection that will be false if 'connection' evaluates to true. In this case an error message is printed and the 'mysql_error()' will be called.
C. The code has a conditional expression '!' (i.e. not) connection that will be true if 'connection' evaluates to false. In this case, the connection will be established.
D. The code has a conditional expression '!' (i.e. not) connection that will be true if 'connection' evaluates to false. In this case an error message is printed and the 'mysql_error()' will be called then the connection will be established.

PART 2
Short questions (20 marks)

Question 1: (4 marks)
a) Discuss the concept of "Warriors of the Net". **(2 marks)**
b) Discuss the website Linear structure, draw a figure to represent Linear structure. **(2 marks)**

Question 2: (4 marks)
a) Discuss the concept of Accessibility and the mobile web. **(2 marks)**
b) Each element in the HTML document tree will generate zero or more boxes, and according to the W3C the layout of boxes is governed by several points, list these points. **(2 marks)**

Question 3: (4 marks)
a) What do we mean by well-formed in the context of XHTML? List four points. **(2 marks)**
b) In reference to forms, discuss CGI and state its main disadvantage. **(2 marks)**

Question 4: (4 marks)
a) Explain four of the Web service operations of SOA. **(2 marks)**
b) Draw a figure that shows the Web service operations of SOA. **(2 marks)**

Question 5: (4 marks)
a) Explain the Multi-tier architecture. **(2 marks)**
b) Explain Cloud architecture. **(2 marks)**

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PART 3

Long questions (30 Marks)

Question 1: (5 marks)

Write HTML code included with a Java script code that will perform the following requests:

a. Ask the user to input two numbers.
b. Calculate and print the average of the two numbers.
c. If the first number is greater than the second number, then display the message: "The first number is the maximum".
d. If the second number is greater than the first number, then display the message: "The second number is the maximum".
e. Otherwise display the message: "Both numbers are equal".

Question 2: (5 marks)

What will be the output if we run the following code?

```
<html>
 <body>
  <script>
   function doit(n1,n2)
    {
     value = 0;
     for (i=n2 ; i >= n1 ;  i= i-2)
      {
       value = value + i;
       document.write("<br> " + value ) ;
      }
    }
   doit(2,10);
</script>
</body>
</html>
```

Question 3: (10 Marks)

Write a PHP code to perform the following:

a) Create two arrays that are having the following details:
   $temp2023 = ("10","20","25")
   $temp2024 = ("12","18","21")
b) Calculate the average of each index of both arrays and store it in a third array named $average then print the $average array.

Note:
The first value of $temp2023 and first value $temp2024 arrays are 10 and 12 so the average is 11 which will be stored as the first value in the $average array. Thus the $average array values will be ("11","19","23").

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Question 4: (10 marks) (1 mark for each request)
Develop a simple web page containing the following specification:

1. Page Title "Data Process Level System".
2. Header 1 "Data Process Level".
3. Form action to locate the script that will process the form data "http//norton.open.ac.uk/reflect.php".
4. Text Input type field for the username.
5. Password type field for the password.
6. List type for the Admin Level.
7. List type for the User Level.
8. Hidden field to post the value "Approved Level".
9. Submit button with Submit name.
10. Reset Button with Reset name.

The web page should look like the following design (Figure 1:



Figure 1

End of questions

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